SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Subject: Programming with Java	**Sem: III AY: 2026-27**
Assignment No: 12	
TITLE: Write a Java program to create and use user-defined packages.	

Problem Statement

Write a Java program to create and use user-defined packages.

Learning Objectives

To create and use user-defined packages for organizing Java programs.

Theory

Packages are used to organize related classes and interfaces into namespaces. User-defined packages improve code organization, reduce naming conflicts, and enhance reusability. They also support access protection and modular application development. Java packages simplify project management by grouping logically related components together. Large-scale software systems extensively use packages.

Exercise

1. Create a Java program by developing user-defined packages for a College Management System. Create separate packages for student and faculty details, and display their information using classes from these packages.

EXPLORER

- ABC
  - CollegeManage...
    - faculty
      - Faculty.class
      - Faculty.java 1
    - student
      - Student.class
      - Student.java 1
    - CollegeManagemen...
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  - mypackage
  - studentpackage
  - LibraryDemo.class
  - LibraryDemo.java
  - MobileDemo.class
  - MobileDemo.java
  - mypackage.java 2

CollegeManagement.java 7 | Student.java 1 | Faculty.java 1

CollegeManagement > CollegeManagement.java > CollegeManagement

```java
import student.Student;
import faculty.Faculty;

public class CollegeManagement
{
    public static void main(String[] args)
    {
        Student s = new Student("Sumit", 195);
        Faculty f = new Faculty("Dr. Sharma", "Java");

        System.out.println("Student Details:");
        s.display();

        System.out.println();

        System.out.println("Faculty Details:");
        f.display();
    }
}
```

CollegeManagement > student > Student.java > Student

```java
package student;

public class Student
{
    String name;
    int rollNo;

    public Student(String name, int rollNo)
    {
        this.name = name;
        this.rollNo = rollNo;
    }

    public void display()
    {
        System.out.println("Student Name: " + name);
        System.out.println("Roll No: " + rollNo);
    }
}
```

CollegeManagement > faculty > Faculty.java > Faculty > name

```java
package faculty;

public class Faculty
{
    String name;
    String subject;

    public Faculty(String name, String subject)
    {
        this.name = name;
        this.subject = subject;
    }

    public void display()
    {
        System.out.println("Faculty Name: " + name);
        System.out.println("Subject: " + subject);
    }
}
```

Output:

```
[Running] cd "/Users/sumit/Documents/java/java1/abc/CollegeManagement/" && javac CollegeManagement.java && java CollegeManagement
Student Details:
Student Name: Sumit
Roll No: 195

Faculty Details:
Faculty Name: Dr. Sharma
Subject: Java

[Done] exited with code=0 in 0.371 seconds
```

2. Create a Java program by developing a **user-defined package** named library that contains a class to store and display book details such as book ID, title, author, and price. Import the package into the main program and display the book information.

```java
import library.Book;

public class LibraryDemo
{
    public static void main(String[] args)
    {
        Book b = new Book(101, "Java Programming", "James Gosling", 500);

        System.out.println("Book Details:");
        b.display();
    }
}
```

```java
package library;

public class Book
{
    int bookId;
    String title;
    String author;
    double price;

    public Book(int bookId, String title, String author, double price)
    {
        this.bookId = bookId;
        this.title = title;
        this.author = author;
        this.price = price;
    }

    public void display()
    {
        System.out.println("Book ID: " + bookId);
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("Price: " + price);
    }
}
```

Output:

```
[Running] cd "/Users/sumit/Documents/java/java1/abc/LibraryDemo/" && javac LibraryDemo.java && java LibraryDemo
Book Details:
Book ID: 101
Title: Java Programming
Author: James Gosling
Price: 500.0

[Done] exited with code=0 in 0.349 seconds
```

Conclusion:

The programs successfully demonstrate the creation and use of user-defined packages in Java. Separate packages were created to organize student, faculty, and book-related classes. The import statement was used to access classes from user-defined packages in the main program. This approach improves code organization, reusability, and maintainability. Thus, user-defined packages provide an effective way to group related classes and manage larger Java programs.